



Modeling tail risks of inflation using unobserved component quantile regressions[☆]

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ABSTRACT

This paper proposes methods for Bayesian inference in time-varying parameter (TVP) quantile regressions (QRs) featuring conditional heteroskedasticity. I use data augmentation schemes to render the model conditionally Gaussian and develop an efficient sampling algorithm. Regularization of the high-dimensional parameter space is achieved via dynamic shrinkage priors. The merits of the proposed approach are illustrated in a simulation study, and a simple version of TVP-QR based on an unobserved components model is applied to dynamically trace the quantiles of inflation in the United States, the United Kingdom and the euro area. In an out-of-sample forecast exercise, I find the proposed model to be competitive and perform particularly well for higher-order and tail forecasts. A detailed analysis of the resulting predictive distributions reveals that they are sometimes skewed and occasionally feature heavy tails.

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1. Introduction

Predictive inference is often concerned with point forecasts of the conditional mean of some economic or financial series. Recent contributions to the forecasting literature stress to take into account higher-order moments of the predictive distribution. Tail risks in particular have been studied extensively in the finance literature, but interest in the tails of distributions of key macroeconomic variables is a comparatively recent phenomenon. In an influential paper, [Adrian et al. \(2019\)](#) find time-varying downside risks in the distribution of GDP growth conditional on economic and financial conditions. [Adams et al. \(2021\)](#) use a similar framework to quantify risks to several other key macroeconomic variables, and come to the same conclusions: risks are time-varying, potentially asymmetric, and partly predictable.¹

A popular approach to model tail risks is the quantile regression (QR). Originally proposed by [Koenker and Bassett \(1978\)](#), QR is designed to estimate the conditional quantiles of some endogenous variable. I add to the literature on QRs featuring time-varying parameters (TVPs) in a unified state space framework. TVPs have proven useful in addressing parameter change

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¹ Other papers addressing risks to macroeconomic variables are [Carriero et al. \(2020a,b\)](#); [Clark et al. \(2021a,b\)](#); [De Nicolò and Lucchetta \(2017\)](#); [Galbraith and van Norden \(2019\)](#); [Giglio et al. \(2016\)](#); [Kiley \(2018\)](#); [Manzan \(2015\)](#); [Plagborg-Møller et al. \(2020\)](#) or [Delle Monache et al. \(2021\)](#).

in macroeconomic and financial series in the context of structural inference, and often improve predictive accuracy when interest centers on forecasting.²

Several previous contributions consider varying coefficient QRs (see [De Rossi and Harvey, 2006](#); [Kim, 2007](#); [Oka and Qu, 2011](#); [Wang et al., 2009](#)); others rely on approximate or nonparametric approaches to account for structural breaks when estimating higher-order moments of the distribution of some series (see [Cai and Stander, 2008](#); [Chen and Gerlach, 2013](#); [Gerlach et al., 2011](#); [Gonçalves et al., 2020](#); [Griffin and Mitrodima, 2022](#); [Lim et al., 2020](#); [Liu, 2016](#); [Smith and Maneesoonthorn, 2018](#); [Taddy and Kottas, 2010](#); [Wu and Zhou, 2017](#)). A recent related paper is [Korobilis et al. \(2021\)](#), who study time-varying inflation risks conditional on a set of macroeconomic and financial indicators in the euro area using a variant of TVP-QR.

The Bayesian QR originates from [Yu and Moyeed \(2001\)](#), who establish a correspondence to classical methods by specifying the likelihood function as an asymmetric Laplace distribution. One caveat of this approach is that it complicates the setup of an efficient Gibbs sampling algorithm. This is due to the conditional posterior distributions of the model parameters being available in well-known form only for specific prior choices. As a solution, [Kozumi and Kobayashi \(2011\)](#) introduce auxiliary variables to represent the asymmetric Laplace distribution as a location-scale mixture. This representation is the point of departure for using conventional Gaussian state space methods within the class of QR models.

I show how to extend the Bayesian QR to feature TVPs by considering several reparameterizations and approximations to facilitate the conditional likelihood. The assumption of a constant scale parameter of the asymmetric Laplace likelihood is relaxed, allowing for heteroskedasticity within quantiles. While QR is itself a method to account for non-Gaussian data features, this extension allows the model to decide whether parameter change is attributed to the quantile function, or whether quantiles feature idiosyncratic stochastic volatility processes. This is a crucial aspect when incorporating TVPs to discriminate signals from noise, and to avoid spurious parameter change as a consequence of large exogenous shocks.

Overfitting concerns are alleviated via adopting dynamic shrinkage priors to regularize the parameter space. Dynamic shrinkage is required when the underlying series or relation between variables exhibits a mixture of periods characterized by stable dynamics, abrupt structural breaks or moderate time-variation. Tail observations occur infrequently by definition, and this property could be particularly useful in the case of TVP-QR.

Forecasts of the conditional mean of inflation, but also its dispersion and tail risks, are important to policy makers in central banks and practitioners in the private sector. Several related approaches using QRs have been proposed to model the full distribution of inflation in autoregressive frameworks or conditional on a set of other macroeconomic indicators (see, e.g., [López-Salido and Loria, 2020](#); [Wolters and Tillmann, 2015](#)). From a forecasting perspective, some papers suggest improvements in predictive accuracy by directly modeling the conditional quantiles of inflation (e.g., [Ghysels et al., 2018](#); [Korobilis, 2017](#); [Korobilis et al., 2021](#); [Manzan and Zerom, 2013](#)). [Korobilis \(2017\)](#), for instance, rules out time-variation but aims to identify promising information sets in constant parameter QR models to forecast inflation and finds this specification to forecast well.

Most of the preceding literature on quantile models of inflation includes explanatory variables based on stylized models such as the Phillips curve and small information sets (e.g., [López-Salido and Loria, 2020](#); [Manzan and Zerom, 2013](#)), or model/variable selection approaches in higher-dimensional data environments (e.g., [Korobilis, 2017](#)). While the proposed methods apply to the general case featuring regressors which is illustrated using synthetic data, I consider a QR-version of a comparatively simple TVP model that has had great success in forecasting the conditional mean of inflation in empirical applications: the unobserved component (UC) model of [Stock and Watson \(2007\)](#). Variants and extensions of this model featuring several additional unobserved factors (e.g., [Chan et al., 2018](#); [2013](#); [Jarociński and Lenza, 2018](#)), or the UC stochastic volatility in mean (UC-SVM) model proposed by [Chan \(2017\)](#) exhibit strong overall forecast performance for inflation.

The probabilistic Gaussian error distributions in such mean-based specifications allow to compute quantiles as a by-product. In this paper, I shed light on the question whether it is beneficial to model the quantiles of inflation explicitly within the class of unobserved components models, and how the resulting shapes of the predictive distributions differ along key dimensions such as skewness or heavy tails. Special emphasis is put on investigating the performance of this approach in different parts of the predictive distribution. The model applied to inflation series from the United States (US), United Kingdom (UK) and the euro area (EA) is an unobserved components quantile regression (UCQR). UCQR is a simple and easy to use yet powerful variant among the UC model class, which allows for economically interpreting the resulting parameter estimates directly as the quantile-based evolution of potentially asymmetric risks to trend inflation.

I compare density and tail forecasts for four model specifications: the UC-SV model, the UC-SVM model, an autoregressive Bayesian QR and the UCQR model. The latter is estimated using several alternative routes for imposing shrinkage on the TVPs to assess empirical properties of the shrinkage priors, with and without time-varying scale parameter, and post-processed in two ways to guarantee monotonic quantile estimates. The results indicate that variants of UCQR perform particularly well for higher-order and tail forecasts. These findings suggest that the proposed models are useful to policy makers in central banks whose key interest is the medium and longer term (i.e., lower-frequency shifts in the distribution of the inflation process). Among the considered UCQR-models, there is some heterogeneity with respect to the considered economies. Conditional heteroskedasticity is required to improve predictive accuracy in the US and the EA, while homoskedasticity is sufficient

² Examples using TVP models for structural inference are [Mumtaz and Theodoridis \(2018\)](#); [Primiceri \(2005\)](#) or [Paul \(2020\)](#). Recent forecasting applications featuring TVP models are, for instance, [Aastveit et al. \(2017\)](#); [D'Agostino et al. \(2013\)](#); [Huber et al. \(2021\)](#) or [Yousuf and Ng \(2021\)](#).

in the UK. The dynamic horseshoe prior for the TVPs exhibits the most consistent performance across forecast metrics, horizons and economies. A detailed analysis of the resulting predictive distributions reveals that they are sometimes skewed and occasionally feature heavy tails.

The rest of the paper is structured as follows. [Section 2](#) presents a general version of TVP-QR, adopts priors for dynamic shrinkage and illustrates the proposed model with an artificial data exercise. [Section 3](#) discusses details about UCQR as a special case of the general framework, and motivates this specification in light of stylized facts about inflation. [Sections 4](#) and [5](#) apply the model to a study of inflation dynamics in the US, UK and EA, and contain in-sample results alongside predictive inference. [Section 6](#) concludes. Supplementary materials including further details, the sampling algorithm and additional empirical results are provided in the Online Appendix.

2. Econometric framework

2.1. Quantile regression with time-varying parameters

Let $\{y_t\}_{t=1}^T$ be a scalar time series and $\{\mathbf{x}_t\}_{t=1}^T$ a $K \times 1$ -vector of explanatory variables at time $t = 1, \dots, T$. The explanatory variables may comprise an intercept, observed/latent factors, additional covariates or lags of the endogenous variable. A general version of TVP-QR is given by:

$$y_t = \mathbf{x}_t' \boldsymbol{\beta}_{pt} + \epsilon_t, \quad \text{with} \quad \int_{-\infty}^0 f_p(\epsilon_t) d\epsilon_t = p. \quad (1)$$

Define $q_p(\mathbf{x}_t) = \mathbf{x}_t' \boldsymbol{\beta}_{pt}$ as the p th quantile regression function of y_t conditional on $\mathbf{x}_t$, for $p \in (0, 1)$. The regression coefficients are collected in a $K \times 1$ -vector $\{\boldsymbol{\beta}_{pt}\}_{t=1}^T$. They vary over time and are specific to the p th quantile. The error term ϵ_t with density $f_p(\bullet)$ has its p th quantile equal to zero. In principle, any error distribution that satisfies this could be used to model the p th conditional quantile. Following [Yu and Moyeed \(2001\)](#), I choose $f_p(\bullet)$ to be the asymmetric Laplace (AL_p) distribution.³ [Kozumi and Kobayashi \(2011\)](#) use a location-scale mixture representation of the AL_p distribution to cast (1) as a conditionally Gaussian model.

In addition to TVPs, I extend the Bayesian QR to feature a time-varying scale parameter similar to a stochastic volatility model.⁴ To achieve this, I define auxiliary variables $v_{pt} \sim \mathcal{E}(\sigma_{pt})$, which follow an exponential distribution with time-varying scaling σ_{pt} , and $u_t \sim \mathcal{N}(0, 1)$. The model in (1) can be written as:

$$y_t = \mathbf{x}_t' \boldsymbol{\beta}_{pt} + \theta_p v_{pt} + \tau_p \sqrt{\sigma_{pt} v_{pt}} u_t, \quad \theta_p = \frac{1-2p}{p(1-p)}, \quad \tau_p^2 = \frac{2}{p(1-p)}. \quad (2)$$

The term $\theta_p v_{pt}$ shifts the location of the observations in y_t , while $\tau_p \sqrt{\sigma_{pt} v_{pt}}$ reweights them such that $\boldsymbol{\beta}_{pt}$ targets the relationship between y_t and $\mathbf{x}_t$ with respect to the p th quantile.⁵ To see this more clearly, let $\tilde{y}_{pt} = (y_t - \theta_p v_{pt}) / (\tau_p \sqrt{\sigma_{pt} v_{pt}})$ and $\tilde{\mathbf{x}}_{pt} = (\tau_p \sqrt{\sigma_{pt} v_{pt}} \mathbf{I}_K)^{-1} \mathbf{x}_t$ with $\mathbf{I}_K$ denoting an identity matrix of size K .

Conditional on v_{pt} and σ_{pt} , [Eq. \(2\)](#) can be written as a standard TVP regression:

$$\tilde{y}_{pt} = \tilde{\mathbf{x}}_{pt}' \boldsymbol{\beta}_{pt} + u_t, \quad u_t \sim \mathcal{N}(0, 1). \quad (3)$$

The time-varying scales σ_{pt} introduce additional flexibility by allowing for both quantile and time-specific differences in volatilities. This allows the model to discriminate between time-varying signals and noise, and induces a varying degree of smoothness in quantile estimates. In other words, this relates to the notion that TVP models must decide whether to capture breaks in the dependent variable as information worth fitting within the conditional quantile, or to consider it as an idiosyncratic shock with drifting volatility that is best captured in the error term. The Online Appendix provides further details on the trade-off between TVPs and the time-varying scale parameter.

I introduce time-variation in the quantile specific regression coefficients and the logarithmic scale parameters via random walk state equations:

$$\boldsymbol{\beta}_{pt} = \boldsymbol{\beta}_{pt-1} + \boldsymbol{\eta}_{pt}, \quad \boldsymbol{\eta}_{pt} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}_{pt}), \quad (4)$$

$$\log(\sigma_{pt}) = \log(\sigma_{pt-1}) + e_{pt}, \quad e_{pt} \sim \mathcal{N}(0, \varsigma_p^2), \quad (5)$$

with $K \times K$ -matrix $\boldsymbol{\Omega}_{pt} = \text{diag}(\omega_{p1,t}, \dots, \omega_{pK,t})$ collecting independent state innovation variances on its diagonal and ς_p^2 referring to the state innovation variance of the scale parameters. Note that combining (3) and (4) yields a state space model with standard normal measurement errors. This enables the use of all standard methods for Gaussian state space models that are available in frameworks specified in terms of the conditional mean.

³ This is due to the correspondence between frequentist and Bayesian inference that this likelihood implies. The error term with respect to quantile p is $\epsilon_t \sim \text{AL}_p(\sigma_{pt})$, with density $f_p(\epsilon_t) = p(1-p)/\sigma_{pt} \exp(-\rho_p(\epsilon_t)/\sigma_{pt})$ where $\rho_p(x) = x(p - \mathbb{I}(x < 0))$ is the check/loss function and $\mathbb{I}(\bullet)$ is an indicator function which yields one if its argument is true and zero otherwise.

⁴ Other approaches to introduce conditional heteroskedasticity within quantiles are GARCH-type models (e.g., [Caporin et al., 2018](#); [Gerlach et al., 2011](#)).

⁵ Note that the product $\sqrt{\sigma_{pt} v_{pt}}$ arises solely due to the alternative representation of the AL_p distribution, and poses no concerns due to v_{pt} being independent over time and $\log(\sigma_{pt})$ following an autoregressive law of motion, alongside the presence of v_{pt} in the “mean” of the conditional model.

2.1.1. Priors for the time-varying coefficients

Time-variation for the k th coefficient in β_{pt} is governed by $\omega_{pk,t}$, for $k = 1, \dots, K$. There are several options to model these variances, each of them being consequential for the dynamic evolution of the states. The default option is to disregard varying degrees of time variation and rely on a constant specification, that is, $\omega_{pk,1} = \dots = \omega_{pk,T} = \omega_{pk}$. A common prior in this case is the *inverse Gamma* distribution (subsequently labeled iG), such that $\omega_{pk} \sim \mathcal{G}^{-1}(m, n)$. In the empirical application, I choose a weakly informative prior with $m = n = 0.1$.

To impose a time-varying degree of shrinkage, the first option is to construct a prior which detects the necessity of time variation on a t -by- t basis. The prior I assume is $\omega_{pk,t} = \lambda_{pk}^2 \phi_{pk,t}^2$, with $\lambda_{pk} \sim \mathcal{C}^+(0, 1)$ and $\phi_{pk,t} \sim \mathcal{C}^+(0, 1)$. $\mathcal{C}^+$ refers to the half-Cauchy distribution. This consideration implies that there is no specific state evolution for $\omega_{pk,t}$, and for this reason I refer to it as the *static horseshoe* prior (labeled shs). In other words, I impose an overall degree of shrinkage λ_{pk} towards constancy, with scalings $\phi_{pk,t}$ providing local adaptiveness for periods where shifts in the model parameters are required. This prior is closely related to those used in Korobilis et al. (2021) or Prüser (2021).

Persistence in the shrinkage process can be achieved by the *dynamic horseshoe* prior (labeled dhs). Here, I assume that $\omega_{pk,t} = \lambda_{p0}^2 \lambda_{pk}^2 \phi_{pk,t}^2$ and consider this quantity on the log-scale to obtain a joint law of motion for $\psi_{pk,t} = \log(\omega_{pk,t}) = \log(\lambda_{p0}^2 \lambda_{pk}^2 \phi_{pk,t}^2)$,

$$\psi_{pk,t} = \mu_{\psi,pk} + \varphi_{pk}(\psi_{pk,t-1} - \mu_{\psi,pk}) + v_{pk,t}, \quad v_{pk,t} \sim \mathcal{Z}(c, d, 0, 1). \quad (6)$$

Following Kowal et al. (2019), this representation establishes a dynamic version of the horseshoe prior (for $c = d = 1/2$), where λ_{p0} acts as global shrinkage parameter specific to quantile p , λ_{pk} is a covariate specific shrinkage parameter, $\phi_{pk,t}$ a covariate and time-specific shrinkage parameter and $\mathcal{Z}$ denotes the Z-distribution. The priors on the shrinkage parameters are $\lambda_{p0} \sim \mathcal{C}^+(0, 1/(TK))$, $\lambda_{pk} \sim \mathcal{C}^+(0, 1/TK)$ and $\lambda_{pk} \sim \mathcal{C}^+(0, 1)$.

2.1.2. Other priors

The prior setup is completed by assuming an inverse Gamma prior with $\sigma_p \sim \mathcal{G}^{-1}(a/2, b/2)$ for the case of a time-invariant scale parameter (labeled TIS) of the AL_p distribution (i.e., for $\sigma_{p1} = \dots = \sigma_{pT} = \sigma_p$), and an inverse gamma prior on the state innovation variance of the logarithmic time-varying process of the scale parameter (labeled TVS), $\varsigma_p^2 \sim \mathcal{G}^{-1}(e, f)$. For the empirical application, I set $a = b = 0.1$, $e = 3$ and $f = 0.3$ to achieve weakly informative priors.

The resulting posterior distributions and details on the sampling algorithm are provided in the Online Appendix. Disregarding a number of draws as burn-in, the Markov chain Monte Carlo (MCMC) algorithm – which involves blocks of Gibbs sampling, forward-filtering backward-sampling (FFBS, Carter and Kohn, 1994; Frühwirth-Schnatter, 1994) for the TVPs, and a random walk Metropolis Hastings algorithm similar to Jacquier et al. (2002) for the time-varying scales – delivers draws from the desired posterior distributions. In the empirical application of this paper, I discard the initial 3000 draws as burnin and use each third of the 9000 subsequent draws for posterior and predictive inference.

2.2. Parallelization and noncrossing quantiles

Note that there is no dependence between coefficients across quantiles for QRs as in (1), so estimation for different values of p can easily be parallelized.⁶ This feature enables the individual quantile-specific models to be estimated roughly in the same time as a model targeting the unconditional mean. However, this comes with a caveat. Disjoint models of quantiles may be problematic, since independent estimates of the quantile function are not guaranteed to be monotonic, i.e., $q_{p_1}(y_t | \mathbf{x}_t) < q_{p_2}(y_t | \mathbf{x}_t)$ for $p_1 < p_2$.⁷

To ensure monotonicity I rely on an auxiliary Gaussian process regression (GPR, see Williams and Rasmussen, 2006) and post-process the collected MCMC draws for all considered quantiles using the two-step approach discussed in Rodrigues and Fan (2017). Let $p = \{0.05, 0.10, \dots, 0.90, 0.95\}$ denote $P = 19$ quantiles of interest, and define $\mathbf{p} = (0.05, 0.10, \dots, 0.90, 0.95)'$.

The approach of Rodrigues and Fan (2017) involves constructing a $P \times P$ matrix

$$\begin{aligned} \mathbf{Q}_{t,(p,\tilde{p})}(y_t | \mathbf{x}_t) &= F^{-1}(\tilde{p}; \mathbf{x}_t' \hat{\beta}_{pt}, \hat{\sigma}_{pt}, p) \\ &= \begin{cases} \mathbf{x}_t' \hat{\beta}_{pt} + \frac{\hat{\sigma}_{pt}}{1-p} \log\left(\frac{\tilde{p}}{p}\right), & \text{if } 0 \leq \tilde{p} \leq p, \\ \mathbf{x}_t' \hat{\beta}_{pt} - \frac{\hat{\sigma}_{pt}}{p} \log\left(\frac{1-\tilde{p}}{1-p}\right), & \text{if } p \leq \tilde{p} \leq 1. \end{cases} \end{aligned}$$

This matrix collects induced quantiles, that is, the quantile function $F^{-1}(\bullet)$ of the fitted AL_p model with respect to the full grid of quantiles $\tilde{p} = \{0.05, 0.10, \dots, 0.90, 0.95\}$. The variables $\hat{\beta}_{pt}$ and $\hat{\sigma}_{pt}$ refer to the posterior mean estimates of the respective quantities. Consequently, this auxiliary model provides $P - 1$ additional estimates of the quantile for each p , since the raw quantile estimates $\mathbf{x}_t' \hat{\beta}_{pt}$ are collected on the diagonal of $\mathbf{Q}_{t,(p,\tilde{p})}(y_t | \mathbf{x}_t)$.

⁶ The recursive forecast exercise shown in this paper, for instance, employs a high-performance cluster which distributes all individual quantile regressions across nodes and collects them afterwards. The associated R-code also provides routines to distribute the computational burden to any number of available CPUs on a single machine.

⁷ The requirement of monotonicity follows from the definition of quantiles. Note that Griffin and Mitrodimia (2022) or Wu and Narisetty (2021) propose frameworks to model a set of quantiles in a joint econometric framework. See also Chernozhukov et al. (2010).

This additional information to unify independent quantile regressions is exploited via the following GPR:

$$\mathbf{Q}_{t,(\bullet,\tilde{p})}(y_t|\mathbf{x}_t) = g_t(\mathbf{p}) + \varepsilon_t, \quad g_t(\mathbf{p}) \sim \mathcal{GP}(\mathbf{0}, \mathbf{Y}_t), \quad \varepsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_t).$$

The covariance matrix $\mathbf{\Sigma}_t$ is diagonal and features the posterior variances of the corresponding element $\mathbf{Q}_{t,(\bullet,\tilde{p})}(y_t|\mathbf{x}_t)$ normalized by the number of retained MCMC draws. The Gaussian process covariance is chosen such that it reflects a decreasing function of the distance between quantiles via a squared exponential kernel. The elements of the square matrix $\mathbf{Y}_t$, $v_{t,(p,\tilde{p})}$, are thus defined as

$$v_{t,(p,\tilde{p})} = s^2 \exp\left(-\frac{1}{2w_t^2}(p - \tilde{p})^2\right),$$

where w_t is the bandwidth and s^2 is the (unconditional) variance of the prior. The variance is set to $s^2 = 100$ to yield a comparatively uninformative prior. The final quantile estimate is given by the fitted values of the GPR at all desired levels of p . This corresponds to a weighted average of the underlying induced quantiles, which is consistent and exhibits favorable empirical properties.

It remains to discuss the choice of the bandwidth w_t . [Rodrigues and Fan \(2017\)](#) show that there always exists a bandwidth which guarantees noncrossing quantiles, and note that as $w_t \rightarrow \infty$ one obtains equal weights for the induced quantile, while the case $w_t \rightarrow 0$ puts non-zero weights only on the raw quantile estimates. Consequently, I select the minimal w_t that results in noncrossing quantiles for all p and $t = 1, \dots, T$. Algorithmically, I achieve this by starting with a small value of w_t which is increased iteratively until a non-crossing solution is found. This approach is subsequently referred to as GPt. An alternative is to choose a $w_t = w$ for all t , which is the default setup for constant parameter QRs. The latter choice is labeled GP and typically results in smoother estimates of the quantiles for TVP-QR over time, since w will be the maximum value of the set of minimal values of $\{w_t\}_{t=1}^T$ that guarantee noncrossing quantiles over time. Note that this procedure is applicable to both in-sample quantile estimates and predictions.

2.3. Simulation-based evidence

To illustrate the proposed framework and assess the properties of the algorithm, this subsection provides a simulation study based on artificial data. The baseline data generating process (DGP) is given by:

$$y_t = \sum_{k=1}^K x_{kt} \beta_{kt} + e_t.$$

I simulate $K = 3$ independent predictors x_{kt} , for $t = 1, \dots, 250$ observations. Here, it suffices to note that the DGPs are differentiated both respect to the laws of motion of the β_{kt} 's and how the error term e_t is specified. To assess the performance of the shrinkage priors, I consider gradually evolving coefficients (labeled G in the figures), coefficients featuring large jumps (J) and parameters with persistently evolving state innovation variances (TV).

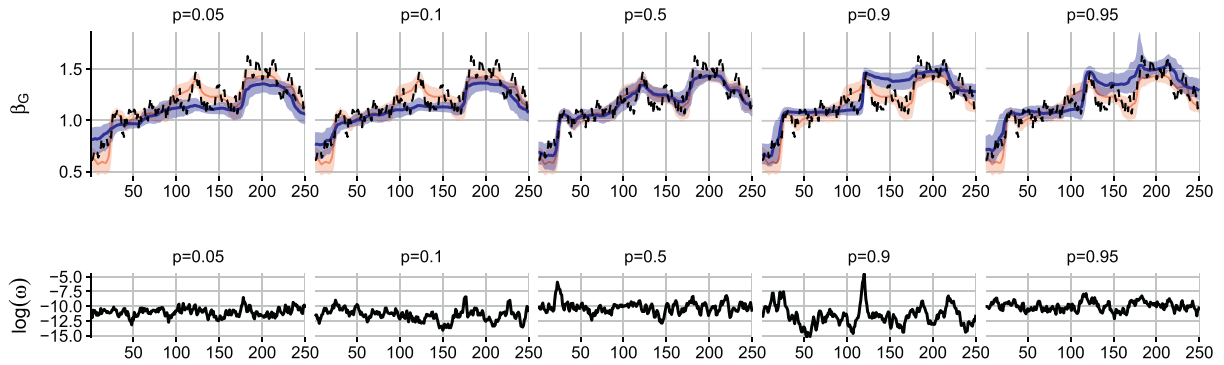
For the error terms, I consider Gaussian errors with constant variance (hom-n) and stochastic volatility (sv), alongside a skew-t distribution (sk-t). To induce heterogeneity in quantile-specific coefficients, I also consider variants where the error term is interacted with x_{kt} , such that I obtain DGPs with proportionally quantile-varying TVPs (het-n), and heterogeneously quantile-varying TVPs (het-t). Details on exact numbers used for simulating the artificial data and how to obtain the true values for quantile-varying coefficients are provided in the Online Appendix.

[Fig. 1](#) shows the path for true parameter values (dashed black lines) for a single realization of the TVPs from a DGP with het-t errors. This specific error distribution is chosen for illustrative purposes since it induces heterogeneity in the coefficient paths across quantiles. Blue shaded areas and lines mark the 68 percent posterior credible sets and median for quantile-specific estimates from TVP-QR-TIS. Orange areas and lines display corresponding parameter estimates from a conventional TVP regression with Gaussian errors featuring stochastic volatility (subsequently referred to as TVP-SV).⁸ TVP-QR is estimated with the dynamic horseshoe prior on the TVPs, as is the mean-based TVP regression to enable one-to-one comparisons. The respective lower panels per coefficient plot the log-posterior median of the shrinkage process captured by the diagonal elements of $\mathbf{\Omega}_{pt}$ for TVP-QR.

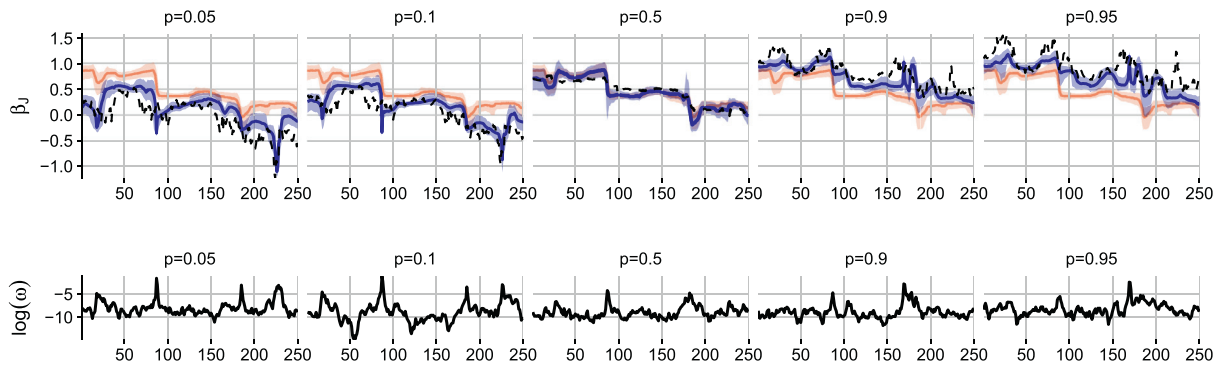
Panels (a) and (c) of [Fig. 1](#) indicate the cases of homogeneous TVPs across quantiles. These charts show that TVP-SV is a very flexible model capable of addressing complex patterns of heteroskedasticity in the error term. Comparing the estimated paths of the coefficients for $p = 0.5$, TVP-QR and TVP-SV exhibit very similar estimates of the underlying regression coefficient. There is some heterogeneity in parameter estimates across quantiles for TVP-QR, which is to be expected given that each of them features its own model. Overall, the estimated paths look reasonable, and TVP-QR recovers the true path of the coefficients remarkably well. The shrinkage process governed by the dynamic horseshoe clearly reflects larger shifts in the coefficients with looser innovation variances, and is capable of extracting signals across all quantiles.

⁸ For TVP-SV, the mean-relationship between y_t and $\mathbf{x}_t$ assumed by the Gaussian errors is symmetric and representative for the whole distribution of y_t . Thus, the implied coefficients for quantiles based on TVP-SV correspond to the estimated path of the single parameter vector specific to the mean, and this estimate is by construction homogeneous across quantiles.

(a) Gradually adjusting coefficient



(b) Coefficient with jumps (and quantile heterogeneity)



(c) Time-varying state innovation variance

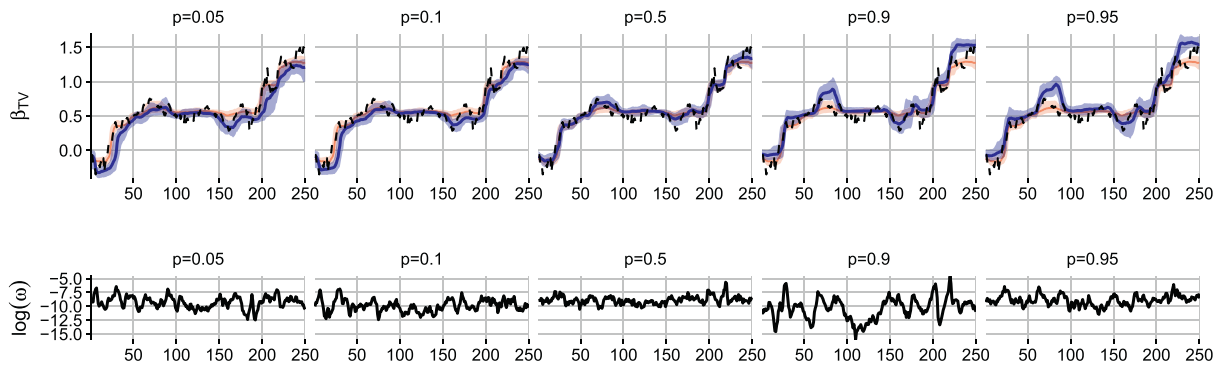


Fig. 1. Parameter estimates for a realization of the TVPs from a DGP with het- t errors. *Notes:* The dashed black lines show the true path of the coefficients. Blue shaded areas and lines mark the 68 percent posterior credible sets and median estimates for quantile-specific TVPs. Orange areas and lines display corresponding parameter estimates from a conventional Gaussian TVP regression with stochastic volatility estimated using the dynamic horseshoe prior (parameter estimates are shown in all quantile-panels for reference). TVP-QR is estimated with the dynamic horseshoe prior on the TVPs and TIS, as is the mean-based TVP regression to enable one-to-one comparisons. The respective lower panels per coefficient plot the posterior median of the shrinkage process captured in Ω_{pt} for TVP-QR on the log-scale. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

The arguably most interesting case is provided in panel (b). Here, the true coefficients vary over time and are heterogeneous with respect to individual quantiles. By construction, the mean-based TVP-SV estimates only one path of this regression coefficient, which implicitly is homogeneous across quantiles. TVP-QR, by contrast, is capable of extracting idiosyncratic parameter movements specific to the quantiles. The corresponding shrinkage processes clearly indicate when large jumps in the coefficients occur. Some imprecisions are visible in the extreme tails, but overall, the estimated path of the coefficient looks satisfactory across all indicated quantiles, as do the paths of the state innovation variances.

Table 1
Mean absolute deviation (MAD) from true coefficients.

Prior	TIS (σ_p)				TVS (σ_{pt})		
	iG	shs	dhs		iG	shs	dhs
<i>Quantile</i>				e_t : hom-n			
$p = 0.05$	1.94	1.53	1.47		3.46	1.97	2.04
$p = 0.10$	1.84	1.26	1.24		2.66	1.57	1.60
$p = 0.50$	1.63	1.02	1.01		1.87	1.17	1.19
$p = 0.90$	1.86	1.23	1.18		2.66	1.51	1.56
$p = 0.95$	1.91	1.53	1.38		3.38	1.85	2.00
				e_t : sv			
$p = 0.05$	2.43	2.18	1.91		3.79	2.24	2.35
$p = 0.10$	2.16	1.70	1.63		2.90	1.73	1.77
$p = 0.50$	1.74	1.04	1.05		2.00	1.12	1.15
$p = 0.90$	2.30	1.67	1.68		2.96	1.86	1.82
$p = 0.95$	2.57	2.20	1.92		3.84	2.31	2.30
				e_t : sk-t			
$p = 0.05$	1.62	1.39	1.18		2.67	1.56	1.59
$p = 0.10$	1.45	1.14	0.99		2.08	1.21	1.22
$p = 0.50$	0.98	0.61	0.63		1.30	0.75	0.76
$p = 0.90$	0.92	0.57	0.55		1.87	1.00	1.03
$p = 0.95$	0.93	0.62	0.58		2.49	1.33	1.34
				e_t : het-n			
$p = 0.05$	1.01	1.01	0.78		1.84	1.04	1.10
$p = 0.10$	0.88	0.68	0.61		1.42	0.83	0.88
$p = 0.50$	0.70	0.48	0.45		0.95	0.56	0.57
$p = 0.90$	0.88	0.69	0.61		1.41	0.82	0.83
$p = 0.95$	1.01	1.01	0.80		1.81	1.05	1.04
				e_t : het-t			
$p = 0.05$	1.02	1.03	0.84		1.75	1.10	1.11
$p = 0.10$	0.90	0.71	0.64		1.35	0.83	0.86
$p = 0.50$	0.67	0.44	0.42		0.86	0.50	0.50
$p = 0.90$	0.90	0.72	0.64		1.33	0.82	0.83
$p = 0.95$	1.02	1.04	0.87		1.77	1.10	1.08

Notes: Dynamic horseshoe (dhs), static horseshoe (shs), inverse Gamma (iG) prior; time-invariant scale (TIS), time-varying scale (TVS) parameter. Blocks of rows refer to the specification of the errors e_t across DGPs. Details on how the underlying data is simulated are provided in the Online Appendix. MAD measures are shown relative to a TVP regression with Gaussian errors and stochastic volatility, which is equipped with a dynamic horseshoe prior. Lower values indicate that TVP-QR produces more accurate parameter estimates.

The previous discussion refers to a single realization of the DGP. Below, I consider a comprehensive Monte Carlo exercise to evaluate how well the variants of TVP-QR are able to recover the true coefficient paths for all DGPs. In particular, I simulate 25 different realizations from all possible DGPs variants and estimate all variants of TVP-QR (differentiated with respect to the prior on the TVPs and the scale parameter) alongside TVP-SV. Let $\hat{\beta}_{pk,t}^{(i,\mathcal{M})}$ denote the i th MCMC draw from the posterior of the quantile-specific coefficient of model $\mathcal{M}$. To assess estimation accuracy, I compute a mean absolute deviation (MAD) measure with the following loss function:

$$\text{MAD}_{\mathcal{M}p} = \frac{1}{I} \sum_{i=1}^I \left(\frac{1}{T} \sum_{t=1}^T \left(\frac{1}{K} \sum_{k=1}^K \left| \beta_{pk,t} - \hat{\beta}_{pk,t}^{(i,\mathcal{M})} \right| \right) \right),$$

where I denotes the total number of MCMC draws and $\beta_{pk,t}$ refers to the true parameter value. Computation draw-by-draw, rather than using the posterior mean/median estimate, offers the possibility to also evaluate how precisely the true coefficient value is recovered. Table 1 shows the mean of $\text{MAD}_{\mathcal{M}p}$ over 25 realizations for each DGP across all models and the quantiles $p \in \{0.05, 0.1, 0.5, 0.9, 0.95\}$ as ratios relative to $\text{MAD}_{\text{TVP-SV}}$. Lower values thus signal that TVP-QR provides more accurate estimates of the true TVPs per quantile than TVP-SV and vice versa.

For DGPs with Gaussian errors, TVP-SV results in more accurate parameter estimates than TVP-QR. This is to be expected, since TVP-QR is less efficient in the sense that it overfits the true DGP relative to the conventional TVP regression by having an individual model for each quantile. Larger relative MAD values can in part be attributed to larger sampling uncertainty surrounding the estimated paths of the quantile specific coefficients. In fact, TVP-SV is specified almost exactly according to the DGP, and it is thus perfectly equipped to recover the true coefficients. For non-Gaussian error distributions, particularly the TIS variants of TVP-QR with a dynamic horseshoe result in substantially more accurate parameter estimates than TVP-SV across all considered quantiles, with some differences depending on the lower versus upper quantile estimates.

In terms of priors, the dynamic horseshoe on average exhibits the most favorable MAD metrics across quantiles and variants of the DGPs. Particularly in the tails, when compared to the static horseshoe, it pays off to consider an autoregressive shrinkage process rather than informing the state innovation variances t -by- t . Without regularization of the TVPs,

reflected in the numbers for the inverse Gamma prior, parameter estimates turn out to be very noisy, which is deleterious for the MAD measures resulting from this prior specification. The TVP-QR variants with TVS perform slightly worse than TIS on average. This finding can be explained by the fact that the flexibility provided by this additional time-varying process widens posterior credible sets, again translating mechanically to larger values in the proposed accuracy metric. However, differentials particularly for the dynamic horseshoe prior are often small in magnitude. It is also worth mentioning that it is not feasible to simulate data from the DGP implied by (1) for TVS. The Monte Carlo exercise is thus by construction biased against this specification.

Summarizing, TVP-QR does a good job of recovering the true paths of the parameter vectors across many different error distributions and for multiple laws of motion of these coefficients. Particularly when the errors in the DGP are non-Gaussian – and there is heterogeneity in coefficients across quantiles – improvements in estimation accuracy are sizeable.

3. The unobserved components quantile regression

This section first provides descriptive statistics and stylized facts about inflation dynamics, which subsequently motivate specifics about model specification in the empirical part of this paper. For the US and the UK, I use the consumer price index starting 1948:Q1 (US) and 1960:Q1 (UK). Both series are available at fred.stlouisfed.org. For the EA, I rely on the harmonized index of consumer prices starting 1990:Q1, available at sdw.ecb.europa.eu. All three range until 2020:Q4. Inflation is defined as $\pi_t = 400 \log(\Pi_t / \Pi_{t-1})$, where Π_t denotes the price index at time t . Fig. 2 shows inflation for all three economies, a histogram and smoothed density estimate of all observations over time, as well as quantiles estimated with a rolling window of 20 observations.

Panel (a) of Fig. 2 indicates π_t for the US. The majority of observations above the 90th and 95th percentile of the unconditional distribution over time occur during the late 1970s and early 1980s. During the 1950s, a period of high volatility, I observe several values below the 10th and 5th percentile. The majority of the remaining observations allocated in the interval $\pi_{0.05} \leq \pi_t < \pi_{0.10}$ occur during or just after recessions until the early 1960s. After the Volcker chairmanship of the Federal Reserve ended in 1987, all observations for inflation in the tails of the unconditional distribution are located below the 10th percentile. This suggests a shift from upside risks of excessive inflation during the 1970s and 1980s towards downside risks of deflation in later periods of the sample. This stylized fact may be linked to the reversal of the relationship between inflation and inflation volatility discussed in Chan (2017). Considering the density plot on the right-hand side, the dynamic evolution of inflation translates to a slightly right-skewed unconditional distribution with heavy tails. Investigating the rolling window quantile estimates suggests substantial movements and asymmetries in quantiles, manifesting in a varying degree of persistence.

Turning to the UK in panel (b) of Fig. 2, the density plot on the right-hand side shows a substantially right-skewed unconditional distribution. All of these observations in the right tail occur between 1975 and 1990, a high-volatility period with substantial upside risks of excessive inflation. The peak is reached just before the brief recession in late 1975. The volatility of inflation in the UK visibly decreased after the European exchange-rate mechanism (ERM) crisis in 1992. Since then, the only observations in the tails of the distributions are those below the 10th and 5th percentile. This shift from upside risk towards downside risk is similar to the case of the US, and may be linked to the Bank of England adopting inflation targeting. Asymmetries in rolling window quantile estimates occur mainly during the late 1970s to the early 1990s, while movements in quantiles are muted from the ERM crisis onwards.

A different pattern is visible for the shorter inflation series of the EA, displayed in panel (c) of Fig. 2. Considering the unconditional distribution of inflation over time on the right-hand side, I observe a slightly left-skewed distribution. The median is approximately located at the European Central Bank's target. The maximum value is observed in the early 1990s, with some values exceeding the 90th percentile in the buildup to the Great Recession, and the rebound between the Great Recession and the European sovereign debt crisis. In the years after the latter crisis, several observations fall into the left tail of the distribution. Similarly deflationary quarters are present during the Great Recession, and in early 2020, at the onset of the recession induced by the COVID-19 pandemic. The rolling window estimate of quantiles suggests only a minor relevance of asymmetries, but indicates that the median of inflation appears to decline. Some periods, such as during the Great Recession and the European sovereign debt crisis, feature markedly wider distributions.

Summarizing, Fig. 2 provides a rough gauge of time-variation in the conditional quantiles of inflation. Movements in quantile estimates based on a rolling window clearly indicate the presence of time-variation across all parts of the conditional distribution. And these patterns are sometimes asymmetric, with movements differing in terms of sign, magnitude, timing and persistence. This observation motivates the choice of imposing dynamic shrinkage. Related priors are designed to allow for sudden breaks in the targeted series, they are equipped to rule out time-variation if dynamics are stable, and they also allow for moderate shifts or trending behavior. These features are clearly desirable for the case of inflation, particularly in the US and the UK, with distinct differences in dynamics observable between periods such as the Volcker Disinflation or the Great Moderation.

The proposed model specification of UCQR is motivated based on this brief discussion and the literature on forecasting the conditional mean of inflation. Popular and successful approaches, particularly for forecasting, are variants of the unobserved components model with stochastic volatility (see, e.g., Chan, 2017; Chan et al., 2018; 2013; Jarociński and Lenza, 2018; Stock and Watson, 2007). The simplest case of this model assumes a persistent (unobserved/latent) trend for the mean, usually augmented with some form of conditional heteroskedasticity. Such models are aimed at providing an accurate model for

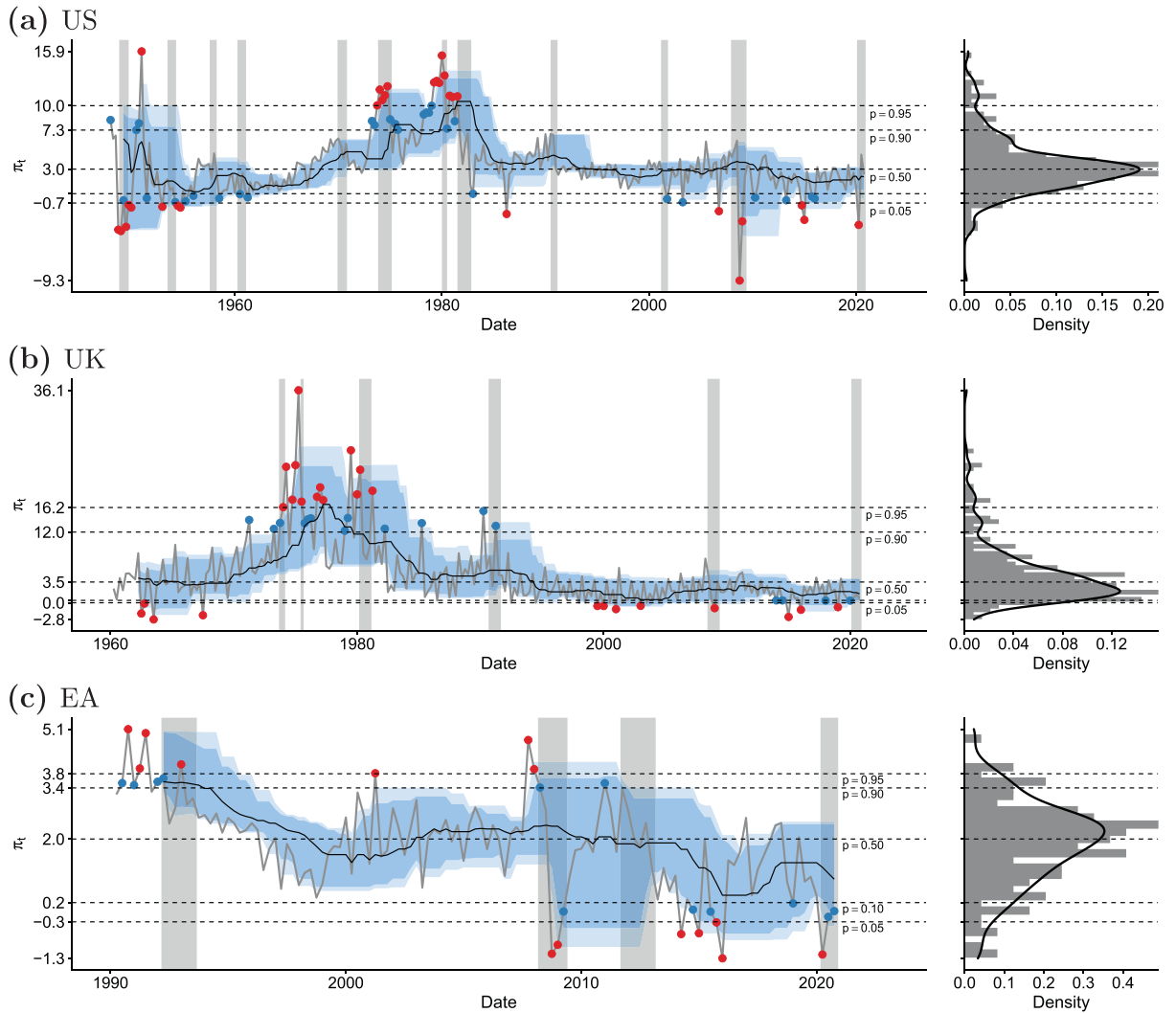


Fig. 2. Inflation series, distribution over time and tail observations. *Notes:* Inflation for the United States (US), the United Kingdom (UK) and the euro area (EA) and histogram/smoothed kernel density estimate of observations. The grey shaded areas are recessions. Black dashed horizontal lines mark the quantiles $p \in \{0.05, 0.10, 0.50, 0.90, 0.95\}$, blue shaded areas (dark blue: $p = 0.10, 0.90$; light blue: $p = 0.05, 0.95$) and solid black line ($p = 0.50$) indicate p estimated with a rolling window of 20 observations (initialized using 8 observations). Red (• with $\pi_t < \pi_{0.05}$ or $\pi_t > \pi_{0.95}$) and blue (• with $\pi_{0.05} \leq \pi_t < \pi_{0.10}$ or $\pi_{0.90} \leq \pi_t < \pi_{0.95}$) dots mark observations in the tails of the distribution, with π_p denoting the p th unconditional quantile over time. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

the conditional mean, which combined with a potentially time-varying variance yields estimates of the conditional quantiles of inflation as a by-product.

In this paper, I ask how explicit models of unobserved components of quantiles compare to mean-based specifications. Interest centers both on overall forecast performance, but also how the competing models perform for different parts of the distribution. The proposed econometric framework thus extends the conventional unobserved components model for the conditional mean by introducing analogous but explicit unobserved components for different quantiles of the distribution of inflation. These estimated quantiles can be interpreted as medium- to long-run measures of inflation risk, since they are quantile-based measures of trend inflation and thus more robust to outliers or other high-frequency fluctuations.

The UCQR model is in fact chosen for several reasons. First, it is very simple and easy to use in the sense that the only required input is the dependent variable. Based on the resulting comparatively small parameter space, it is also very fast computationally, and the quantile-specific intercept terms can directly be interpreted as long-run trend inflation (risk) by quantile. Second, as stated above, it is the quantile-based extension of the UC model of [Stock and Watson \(2007\)](#), which is used extensively for inflation forecasting. A direct comparison to the proposed approach, with individual quantile-models tailored to specific parts of the distribution, appears to be a worthwhile endeavour. As will be discussed below, UC-SV typically results in predictive distributions which are symmetric around the mean, while UCQR allows for additional flexibility

in this regard. Third, while classical approaches using QR to model inflation and inflation risk, such as López-Salido and Loria (2020), are capable of modeling time-varying and asymmetric risk, they require carefully selected predictors. It is unclear, however, which predictors should be included, and whether their predictive capability evolves over time (see, e.g., Korobilis, 2017). This argument is pertinent when multiple economies with potentially varying data availability are to be considered. Finally, the model specification allows for computing iterative forecasts. Iterative forecasts have been found to be preferable to direct forecasts theoretically on the grounds of efficiency and empirically in many cases (see also Marcellino et al., 2006).

3.1. Details on model specification

3.1.1. Variants of unobserved components quantile regressions

I propose to model inflation using the simplest variant of the general TVP-QR model proposed in Section 2. This model generalizes the UC-SV model of Stock and Watson (2007) to modeling the conditional quantiles of inflation. The UCQR model is given by:

$$\pi_t = \alpha_{pt} + \epsilon_t, \quad \epsilon_t \sim \text{AL}_p(\sigma_{pt}), \quad (7)$$

$$\alpha_{pt} = \alpha_{pt-1} + \eta_{pt}, \quad \eta_{pt} \sim \mathcal{N}(0, \omega_{pt}), \quad (8)$$

with ϵ_t following an AL_p distribution. I rely on alternatively stating the AL_p distribution using auxiliary variables discussed in detail in Section 2.1. The competing model specifications include both a time-invariant scale (TIS) and a time-varying scale (TVS) version of UCQR.

To assess the robustness of this approach with respect to priors, I consider the dynamic horseshoe (dhs), the static horseshoe (shs) and a conventional inverse Gamma (iG) prior to determine ω_{pt} . The paper also provides a comparison between unprocessed and processed quantile estimates. Unprocessed estimates (i.e., noncrossing of quantiles is not guaranteed) are referred to as raw. They are shown alongside those adjusted using the two-stage GPR procedure discussed in Section 2.2. GP refers to the bandwidth parameter w being fixed over time, GPt indicates a time-varying w_t .

3.1.2. Competing models

The UCQR model is compared to mean-based versions of the UC model and a conventional autoregressive Bayesian QR. In particular, I consider the following models as competing specifications:

- **Unobserved components with stochastic volatility (UC-SV).** This model assumes a measurement equation of the form $\pi_t = \alpha_t + \epsilon_t$, with error term $\epsilon_t \sim \mathcal{N}(0, \exp(h_t))$, a random walk state equation $\alpha_t = \alpha_{t-1} + \sqrt{\omega_{at}}e_{at}$ and log-volatility process $h_t = h_{t-1} + \zeta_h e_{ht}$. The errors of the state equations follow independent standard normal distributions, $e_{st} \sim \mathcal{N}(0, 1)$ for $s \in \{\alpha, h\}$. This is a variant of the model proposed by Stock and Watson (2007), and the natural competitor to UCQR given that it is specified analogously, but targets solely the conditional mean rather than conditional quantiles.
- **Unobserved components with stochastic volatility in mean (UC-SVM).** This model assumes a measurement equation of the form $\pi_t = \alpha_t + \gamma_t \exp(h_t) + \epsilon_t$ with error term $\epsilon_t \sim \mathcal{N}(0, \exp(h_t))$. The state equations are $\alpha_t = \alpha_{t-1} + \sqrt{\omega_{at}}e_{at}$, $\gamma_t = \gamma_{t-1} + \sqrt{\omega_{\gamma t}}e_{\gamma t}$, and for the stochastic volatilities $h_t = h_{t-1} + \zeta_h e_{ht}$, with e_{st} following independent standard normal distributions for $s \in \{\alpha, \gamma, h\}$. This specification was originally used in Chan (2017), and establishes a time-varying correlation between the state and measurement equation. It thus allows for flexible feedback effects between inflation and its volatility, relating to the literature on financial conditions and uncertainty. A multivariate version of this model in the context of tail risks in GDP growth has been used by Caldara et al. (2021).
- **Autoregressive quantile regression (QR-AR).** This model is a conventional Bayesian QR model featuring the first lag of inflation as predictor in addition to an intercept: $\pi_t = \alpha_p + \beta_p \pi_{t-1} + \epsilon_t$, with $\epsilon_t \sim \text{AL}_p(\sigma_p)$. This model is nested in the general formulation of (1). Different to the other competing models, this specification requires computing direct rather than iterative forecasts.

UC-SV and UC-SVM are implemented and estimated as described in Huber and Pfarrhofer (2021) with a dynamic horseshoe prior. An overview of all model specifications and adjustments is provided in Table 2. The forecast exercise thus features three competing models alongside 18 variants of UCQR governed by the respective prior, treatment of the scale parameter and ex post adjustment of quantile estimates.

4. Conditional mean and quantile models for inflation

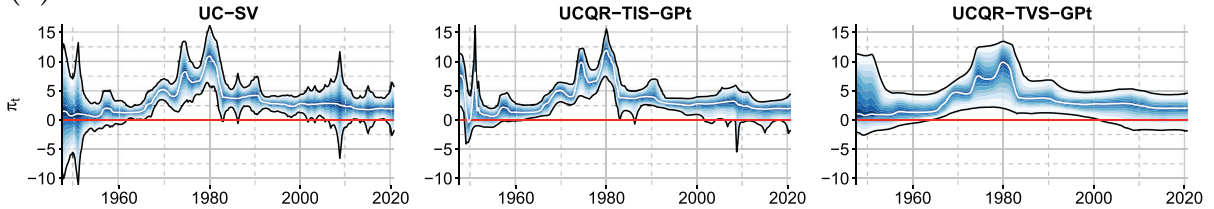
Fig. 3 compares in-sample findings for the US, the UK and the EA. The red line marks zero, the solid black lines are the posterior mean of the 5th and 95th quantiles, and the solid white line is the 50th percentile (median). The blue shaded areas cover quantile pairs (e.g., 10th to 90th percentile) in increments of five. As mentioned above, similar to the unobserved components model based on Gaussian errors, the estimated quantiles are associated with trend inflation, and thus represent medium- to long-run measures of inflation risk by quantile. Variants of UCQR with TIS and TVS, post-processed using a GPR with time-varying bandwidth are shown alongside the conventional UC-SV model. Additional empirical results are provided

Table 2
Model overview.

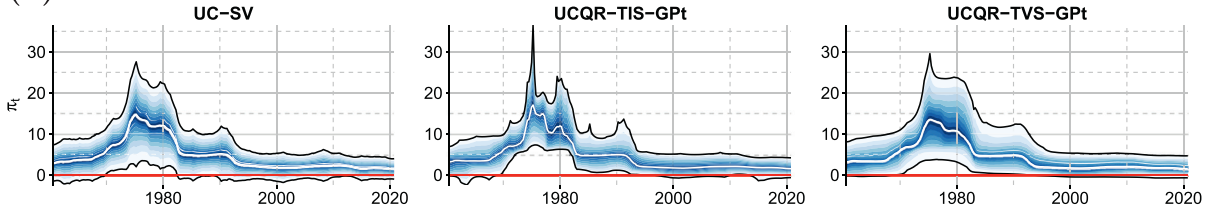
Model	Prior	Volatility/scale	Adjustment
QR-AR	hs	TIS	raw
UC	dhs	SV, SVM	—
UCQR	dhs, shs, iG	TIS, TVS	raw, GP, GPt

Notes: Unobserved components (UC), unobserved components quantile regression (UCQR); horseshoe (hs), dynamic horseshoe (dhs), static horseshoe (shs), inverse Gamma (iG) prior; stochastic volatility (SV), stochastic volatility in mean (SVM), time-invariant scale (TIS), time-varying scale (TVS) parameter; *raw* refers to unprocessed estimates of the conditional quantiles, GP is adjustment using the Gaussian process regression with the bandwidth w fixed over time, GPt marks the Gaussian process regression with time-varying bandwidth w_t .

(a) US



(b) UK



(c) EA

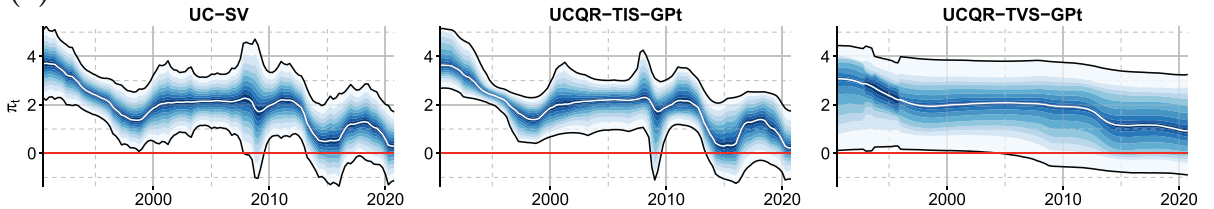


Fig. 3. Unobserved components for UC-SV and variants of UCQR. Notes: Unobserved components model with stochastic volatility (UC-SV) and unobserved components quantile regression (UCQR) with time-invariant (TIS) or time-varying (TVS) scale parameter, estimated with a dynamic horseshoe prior and adjusted using a Gaussian process regression with time-varying bandwidth (GPt). The red line marks zero. The solid black lines are the posterior mean of the 5th and 95th quantiles, the solid white line is the 50th percentile (median). The blue shaded areas cover quantile pairs (e.g., 10th to 90th percentile) in increments of five. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

in the Online Appendix. I consider the posterior mean of the quantile estimates for the UCQR-variants, and rely on MCMC output based on $\pi_t \sim \mathcal{N}(\alpha_t, \exp(h_t))$ for UC-SV.⁹

The charts in Fig. 3 show several interesting differences with respect to the three models and over the cross-section of economies. Most strikingly for the case of the US, UC-SV results in approximately symmetric estimates of upside and downside risk, apparent mainly in the early part of the sample and during the Great Recession. While the mean-based model suggests that both risks are about equally likely, the implied distributions from quantile-based models early in the sample are skewed, and point mainly towards risks of inflation rather than deflation. This situation is mirrored during the Great Recession, with a comparatively stable upper tail of the distribution and elevated risk of deflation for UCQR-TIS-GPt, and a symmetric widening of the distribution in the case of UC-SV. The version of UCQR with TVS overall re-

⁹ In a recent paper, Wu and Narisetty (2021) note that approaches using the AL_p likelihood yield invalid asymptotic variances of posterior distributions, and thus produce incorrect inference from a frequentist perspective. Their findings are of no concern for the purposes of this paper, since both the discussion of in-sample findings and the forecast evaluation metrics require only a point estimate from the posterior distribution of quantiles, and I am not concerned with more intricate inferential problems such as hypothesis testing.

sults in much smoother and persistent estimates of quantiles, particularly for the lower tails during the late 1970s and early 1980s.

Estimates for the quantiles of trend inflation in the UK provide a similar picture compared to the US, albeit with the difference of UC-SV and UCQR-TVS-GPt exhibiting a rather similar dynamic evolution. By contrast, estimates for the case of UCQR-TIS-GPt are much noisier, particularly during the high-inflation period during the late 1970s. The UK provides a prime example to investigate the empirical features of the dynamic horseshoe prior, taking the median as an example. First, a modest degree of time-variation is visible, with a slight upward trend of inflation peaking around 1975. Second, there are several abrupt changes, particularly around 1980 and 1992, with substantial decreases of inflation. These large breaks look similar to break points that one would obtain from running a mixture-innovation or threshold model. Third, there are prolonged periods of virtually no time-variation, e.g., between 1981 and 1990, or after 1995.

For the case of the EA, different to the US and UK, UC-SV and UCQR-TIS-GPt result in rather similar estimates of the quantiles of inflation over time. UCQR-TVS-GPt, however, looks very different to the former two models, with much smoother estimates for all considered quantiles. In the early part of the sample, the implied distribution is much wider, substantially skewed, and movements particularly in the tails are muted. In fact, the lower tail does not exhibit much time-variation at all until around 2005. This again showcases the effects of imposing dynamic shrinkage.

Summarizing, two key findings are worth noting. First, mean-based models typically indicate symmetric risks, while UCQR allows for asymmetric movements in quantiles and thus skewed and heavily tailed distributions. This finding arises from defining distinct quantile regressions with individual loss functions that specifically target different parts of the distribution, rather than average considerations solely based on the conditional mean. In fact, UCQR infers this solely from the target series, without the need for carefully selecting additional predictors to induce such movements as would be the case for conventional QRs. Second, using a time-varying scale parameter for the AL_p distribution usually results in much smoother estimates of the conditional quantiles. Econometrically, this can be explained by noting that larger values of σ_{pt} downweight extreme observations. Relaxing the assumption of conditionally homoskedastic errors in conjunction with a flexible model for the conditional quantiles (of trend inflation) allows the model to decide whether it fits the underlying series aggressively or puts such dynamics in the stochastic error component ϵ_t . Conditional heteroskedasticity clearly captures part of the movements in the underlying quantiles, thereby leaving less information to fit via the time-varying intercept by quantile.

5. Predictive inference

To assess predictive accuracy of the competing models, I conduct a pseudo out-of-sample exercise using an expanding window. The sample is split into training and holdout periods. I estimate the models using data from the first training sample to produce h -step ahead forecasts for $h \in \{1, 4, 12\}$, and evaluate these using the corresponding realized values in the holdout. Subsequently, an additional observation from the holdout is added to the training sample and the procedure is iterated until the holdout is exhausted. The initial sample is chosen such that it consists of 50 quarterly observations for all three economies, resulting in differently sized holdout periods.

5.1. Forecast evaluation

I consider the quantile weighted continuous ranked probability score (CRPS) to measure the overall accuracy of the density forecasts for different parts of the distribution. Let $\pi_{t+h}^{(fp)}$ denote the point forecast of quantile p at time t for horizon h , and $\pi_{t+h}^{(r)}$ is the realization of the inflation measure in the holdout. For UC-SV and UC-SVM, $\pi_{t+h}^{(fp)}$ is the p th quantile of the predictive distribution. For UCQR, I obtain forecasts for the p th quantile in three variants: the unprocessed mean of the distribution for each quantile, the output of the GPR with the bandwidth w fixed for all t (UCQR-GP), and a GPR with w_t selected t -by- t (UCQR-GPt).

I first define the quantile score (QS, see, e.g., [Giacomini and Komunjer, 2005](#); [Gneiting and Raftery, 2007](#)) for quantile p at time t for h -steps ahead, which is computed as:

$$QS_{p,t+h} = (\pi_{t+h}^{(r)} - \pi_{t+h}^{(fp)}) \times (p - \mathbb{I}(\pi_{t+h}^{(r)} < \pi_{t+h}^{(fp)})),$$

where $\mathbb{I}$ is the usual indicator function. Raw values for QSs to assess tail forecast performance at specific quantiles are provided in the Online Appendix. Quantile weighted CRPSs, following [Gneiting and Ranjan \(2011\)](#), are defined as:

$$CRPS_{t+h}(w_p) = \int_0^1 w_p QS_{p,t+h} dp,$$

with non-negative weights w_p on the unit interval putting emphasis on specific parts of the distribution. Five weighting schemes are considered for CRPSs: (a) equal $w_p = 1$, (b) tails $w_p = (2p - 1)^2$, (c) left tail $w_p = (1 - p)^2$, (d) center $w_p = p(1 - p)$, (e) right tail $w_p = p^2$, for $p \in \{0.05, 0.10, \dots, 0.90, 0.95\}$.

[Tables 3–5](#) report the quantile weighted CRPS variants for the US, UK and EA as averages over the holdout. They are benchmarked relative to the UC-SV model in ratios. The row for the benchmark displays raw values, all other models are shown relative to the benchmark. Smaller numbers are better and values below one indicate that the respective model improves upon the benchmark.

Table 3
Density forecast metrics for US inflation.

<i>h</i> =	CRPS														
	None			Tails			Left			Center			Right		
	1	4	12	1	4	12	1	4	12	1	4	12	1	4	12
UC-SV	0.48	0.64	0.81	0.10	0.14	0.18	0.15	0.19	0.24	0.09	0.13	0.16	0.14	0.19	0.25
UC-SVM	0.96	0.98	1.06	0.95	1.00	1.09	0.97	1.01	1.08	0.96	0.98	1.05	0.94	0.96	1.05
QR-AR	1.02	1.01	0.99	1.02	1.02	1.01	1.02	1.00	0.98	1.02	1.00	0.98	1.02	1.02	1.00
UCQR-TIS															
dhs	1.02	1.01	0.99	1.02	1.02	1.01	1.02	1.00	0.98	1.02	1.00	0.98	1.02	1.02	1.00
shs	1.00	1.01	1.01	1.01	1.03	1.04	1.01	1.03	1.03	1.00	1.00	1.01	0.99	1.00	1.01
iG	1.01	1.04	1.08	1.04	1.11	1.15	1.02	1.08	1.11	1.00	1.02	1.06	1.01	1.03	1.07
dhs-GP	1.02	1.01	0.99	1.02	1.02	1.01	1.02	1.00	0.98	1.02	1.00	0.98	1.02	1.02	1.00
shs-GP	1.00	1.01	1.02	1.00	1.03	1.04	1.01	1.03	1.03	0.99	1.00	1.01	0.99	1.00	1.01
iG-GP	1.00	1.01	1.02	1.00	1.03	1.04	1.01	1.03	1.03	0.99	1.00	1.01	0.99	1.00	1.01
dhs-GPt	1.01	1.04	1.08	1.05	1.12	1.14	1.02	1.09	1.12	1.00	1.02	1.06	1.01	1.02	1.07
shs-GPt	1.01	1.04	1.08	1.04	1.10	1.14	1.02	1.08	1.11	1.00	1.02	1.06	1.01	1.02	1.06
iG-GPt	1.02	1.01	0.99	1.02	1.02	1.01	1.02	1.00	0.98	1.02	1.00	0.98	1.02	1.02	1.00
UCQR-TVS															
dhs	1.14	1.01	0.93	1.13	1.00	0.90	1.09	0.97	0.91	1.15	1.01	0.94	1.19	1.05	0.94
shs	1.13	1.02	0.95	1.10	0.99	0.91	1.07	0.98	0.92	1.14	1.03	0.96	1.19	1.05	0.96
iG	1.09	0.98	0.93	1.07	0.95	0.88	1.06	0.95	0.91	1.10	0.99	0.94	1.12	1.00	0.93
dhs-GP	1.14	1.01	0.93	1.13	1.00	0.90	1.09	0.97	0.91	1.15	1.01	0.94	1.20	1.05	0.94
shs-GP	1.12	1.00	0.94	1.10	0.98	0.90	1.07	0.97	0.91	1.12	1.01	0.95	1.16	1.03	0.95
iG-GP	1.12	1.00	0.94	1.10	0.98	0.90	1.07	0.97	0.91	1.12	1.01	0.95	1.16	1.03	0.94
dhs-GPt	1.09	0.98	0.93	1.07	0.95	0.88	1.06	0.95	0.91	1.10	0.99	0.94	1.12	1.00	0.93
shs-GPt	1.10	0.98	0.93	1.08	0.95	0.88	1.07	0.96	0.92	1.11	1.00	0.95	1.13	1.00	0.93
iG-GPt	1.17	1.03	0.95	1.15	1.03	0.93	1.11	0.99	0.93	1.17	1.03	0.96	1.23	1.08	0.97

Notes: For model abbreviations, see Section 3.1 and Table 2. Quantile weighted continuous ranked probability scores (CRPS) for different weighting schemes. The benchmark shows actual values in italics, all other models are shown relative to the benchmark. The best performing model is indicated in bold.

Table 4
Density forecast metrics for UK inflation.

<i>h</i> =	CRPS														
	None			Tails			Left			Center			Right		
	1	4	12	1	4	12	1	4	12	1	4	12	1	4	12
UC-SV	0.93	0.97	1.02	0.19	0.20	0.21	0.26	0.27	0.29	0.18	0.19	0.20	0.30	0.32	0.32
UC-SVM	1.01	1.00	1.03	0.98	0.98	1.05	1.01	1.00	1.07	1.01	1.00	1.03	0.99	0.99	1.00
QR-AR	1.02	1.01	0.99	1.02	1.03	1.02	1.02	1.00	0.99	1.03	1.01	0.98	1.03	1.03	1.01
UCQR-TIS															
dhs	1.00	1.03	1.02	1.00	1.03	1.04	0.99	1.02	1.01	1.00	1.02	1.01	1.01	1.04	1.03
shs	0.99	1.00	1.00	1.00	1.01	1.04	0.98	0.99	1.01	0.99	0.99	0.99	1.00	1.01	1.00
iG	1.03	0.96	0.99	1.07	0.98	1.05	1.01	0.97	1.04	1.02	0.96	0.98	1.06	0.96	0.97
dhs-GP	1.00	1.03	1.02	1.00	1.03	1.04	0.99	1.02	1.01	1.00	1.03	1.01	1.01	1.04	1.03
shs-GP	0.99	1.00	1.00	1.00	1.01	1.04	0.98	0.99	1.01	0.99	0.99	0.99	1.00	1.01	1.00
iG-GP	0.99	1.00	1.00	1.00	1.01	1.04	0.98	0.99	1.01	0.99	0.99	0.99	1.00	1.01	1.00
dhs-GPt	1.04	0.96	0.99	1.08	0.97	1.05	1.01	0.97	1.04	1.02	0.96	0.98	1.07	0.96	0.97
shs-GPt	1.04	0.97	1.00	1.06	0.98	1.04	1.02	0.97	1.05	1.03	0.96	0.99	1.06	0.97	0.97
iG-GPt	1.00	1.03	1.02	1.00	1.03	1.04	0.99	1.02	1.01	1.00	1.03	1.01	1.01	1.04	1.03
UCQR-TVS															
dhs	1.02	1.05	1.00	1.04	1.07	1.03	1.01	1.03	0.96	1.02	1.05	0.99	1.04	1.08	1.04
shs	1.02	1.02	0.98	1.02	1.02	0.99	1.01	1.02	0.97	1.02	1.03	0.98	1.02	1.03	0.99
iG	1.01	1.00	0.96	1.01	1.01	0.98	1.00	0.99	0.95	1.01	1.00	0.96	1.02	1.02	0.98
dhs-GP	1.02	1.05	1.00	1.04	1.07	1.03	1.01	1.03	0.96	1.02	1.05	0.99	1.04	1.08	1.04
shs-GP	1.01	1.03	0.98	1.02	1.03	1.00	1.00	1.01	0.95	1.01	1.02	0.97	1.02	1.04	1.00
iG-GP	1.01	1.03	0.98	1.02	1.03	1.00	1.00	1.01	0.96	1.01	1.02	0.97	1.02	1.04	1.00
dhs-GPt	1.01	1.00	0.96	1.01	1.01	0.99	1.00	0.99	0.95	1.01	1.00	0.96	1.01	1.02	0.98
shs-GPt	1.01	1.00	0.97	1.00	1.00	0.98	1.00	1.00	0.96	1.01	1.00	0.97	1.01	1.00	0.98
iG-GPt	1.02	1.06	1.00	1.05	1.08	1.04	1.01	1.03	0.96	1.02	1.05	0.99	1.04	1.09	1.05

Notes: For model abbreviations, see Section 3.1 and Table 2. Quantile weighted continuous ranked probability scores (CRPS) for different weighting schemes. The benchmark shows actual values in italics, all other models are shown relative to the benchmark. The best performing model is indicated in bold.

Table 5
Density forecast metrics for EA inflation.

<i>h</i> =	CRPS														
	None			Tails			Left			Center			Right		
	1	4	12	1	4	12	1	4	12	1	4	12	1	4	12
UC-SV	<i>0.30</i>	<i>0.37</i>	<i>0.43</i>	<i>0.06</i>	<i>0.08</i>	<i>0.10</i>	<i>0.09</i>	<i>0.12</i>	<i>0.15</i>	<i>0.06</i>	<i>0.07</i>	<i>0.08</i>	<i>0.09</i>	<i>0.11</i>	<i>0.12</i>
UC-SVM	0.97	1.06	1.05	0.96	1.06	1.03	0.99	1.07	1.02	0.98	1.06	1.06	0.95	1.05	1.08
QR-AR	1.04	1.01	0.99	1.02	1.01	1.01	1.02	0.98	0.96	1.04	1.02	0.99	1.05	1.04	1.02
UCQR-TIS															
dhs	1.04	1.03	1.00	1.05	1.06	1.01	1.05	1.05	1.01	1.03	1.02	0.99	1.02	1.01	0.98
shs	1.02	1.05	1.03	1.03	1.09	1.04	1.05	1.07	1.04	1.02	1.04	1.03	1.00	1.04	1.02
iG	1.01	1.06	1.08	1.04	1.13	1.13	1.05	1.08	1.11	1.00	1.04	1.06	0.98	1.05	1.06
dhs-GP	1.04	1.03	1.00	1.05	1.06	1.01	1.06	1.05	1.01	1.04	1.02	0.99	1.03	1.01	0.98
shs-GP	1.02	1.05	1.03	1.03	1.09	1.04	1.05	1.07	1.04	1.02	1.04	1.03	1.00	1.04	1.02
iG-GP	1.02	1.05	1.03	1.03	1.09	1.04	1.05	1.07	1.04	1.02	1.04	1.03	1.00	1.04	1.02
dhs-GPt	1.01	1.06	1.08	1.04	1.13	1.14	1.05	1.09	1.11	1.01	1.04	1.07	0.98	1.06	1.06
shs-GPt	1.01	1.06	1.08	1.05	1.13	1.13	1.05	1.09	1.11	1.00	1.04	1.07	0.98	1.05	1.06
iG-GPt	1.04	1.03	0.99	1.05	1.06	1.01	1.06	1.05	1.01	1.04	1.02	0.99	1.02	1.01	0.98
UCQR-TVS															
dhs	1.11	0.95	0.88	1.10	0.95	0.84	1.13	0.96	0.83	1.11	0.96	0.89	1.09	0.95	0.92
shs	1.11	0.96	0.89	1.11	0.95	0.85	1.14	0.97	0.85	1.12	0.96	0.91	1.09	0.94	0.93
iG	1.08	0.97	0.88	1.06	0.95	0.83	1.08	0.96	0.83	1.08	0.97	0.89	1.06	0.96	0.92
dhs-GP	1.11	0.95	0.88	1.10	0.95	0.84	1.13	0.96	0.83	1.11	0.96	0.89	1.09	0.95	0.93
shs-GP	1.11	0.96	0.88	1.11	0.96	0.84	1.13	0.97	0.84	1.12	0.96	0.90	1.09	0.95	0.93
iG-GP	1.11	0.96	0.88	1.10	0.95	0.84	1.13	0.97	0.84	1.11	0.96	0.89	1.09	0.95	0.92
dhs-GPt	1.07	0.97	0.88	1.06	0.95	0.83	1.08	0.96	0.83	1.08	0.97	0.89	1.06	0.96	0.92
shs-GPt	1.12	1.01	0.91	1.12	1.04	0.89	1.17	1.05	0.90	1.12	1.00	0.91	1.06	0.97	0.90
iG-GPt	1.11	0.96	0.88	1.10	0.96	0.84	1.13	0.97	0.83	1.11	0.96	0.89	1.08	0.95	0.92

Notes: For model abbreviations, see Section 3.1 and Table 2. Quantile weighted continuous ranked probability scores (CRPS) for different weighting schemes. The benchmark shows actual values in italics, all other models are shown relative to the benchmark. The best performing model is indicated in bold.

Investigating the results in detail, UC-SV and UC-SVM appear to be tough benchmarks to improve upon for all three considered economies, at least in the short-term. UC-SVM typically performs slightly better than the conventional UC-SV. The conditional mean-based models are particularly strong for short-horizon forecasts, with most favorable metrics in all parts of the distribution targeted by the quantile weighted CRPSs. In fact, the best performing specification for one-quarter ahead forecasts in terms of CRPSs is UC-SVM in almost all cases across the three economies. QR-AR overall performs quite similarly to the mean-based UC variants, and never outperforms the best performing UCQR-variant for the accuracy metrics considered here.

For multi-step ahead forecasts, the additional distributional flexibility for the quantile-based model tends to pay off in terms of predictive accuracy. This finding is especially pronounced at the three-year ahead horizon. One of the UCQR specifications is identified as the best performing model measured with the CRPS variants for all considered higher-order forecasts, with the exception of one-year ahead forecasts for the US. The quantile weighted versions of CRPSs target different parts of the distribution, and thus allow to investigate where overall gains stem from in more detail. The results indicate that UCQR, again for higher-order forecasts, results in pronounced gains, especially when focusing on either the left, right or both tails via the weighting scheme. This relates to the preceding discussion of interpreting the estimated components per quantile as long-term measures of trend inflation risk.

Within the class of quantile-based models, some differences with respect to the considered economy are apparent. Allowing for conditional heteroskedasticity (TVS) within quantiles results in more accurate forecasts for the US and the EA. In these two economies, it is worth mentioning that the homoskedastic variant (TIS) fails to improve upon the mean-based models in many cases, while TVS shows improvements ranging between 10 to 20 percent lower CRPSs depending on the economy, horizon and weighting scheme. Intuitively, the findings for the US and EA can be explained by noting that TVS results in a less aggressive fit of the quantile function, which hurts predictive accuracy in the short-run by filtering out high-frequency movements, but pays off in the long-run by improvements in capturing low-frequency movements in tail risks of inflation. This is different in the UK, where TIS seems to be better suited to model the predictive distribution of inflation. However, these gains are more modest, and in general, the differences in forecast metrics are muted for the UK.

Differences in predictive accuracy between priors for UCQR are often muted. In many cases one of the two shrinkage priors is identified as the best performing specification. In cases where the iG prior works best, it does so typically at a small margin. It is worth mentioning that the dynamic horseshoe (dhs) performs well most consistently, compared to the static horseshoe (shs). Turning to the consequences of post-processing quantile estimates to achieve a noncrossing quantile solution, I find that using a fixed bandwidth in the GPR usually yields similar results compared to those allowing for time-variation in this parameter. Interestingly, the raw quantile estimates which are not guaranteed to be monotonic exhibit the best forecast performance in a small number of cases. In a nutshell, adjusting quantiles ex post neither helps nor hurts predictive accuracy much. Note that the Online Appendix provides additional results for raw quantile scores. These signal

substantially larger improvements of up to 50 percent for individual quantiles in many cases, particularly for TVS-variants in the US and the EA.

Summarizing, four key findings from this predictive exercise are worth noting. First, variants of UCQR perform particularly well for higher-order and tail forecasts. For one-quarter ahead forecasts, the mean-based competing models often indicate better performance, but only at small margins and UCQR is competitive. Second, there is some heterogeneity with respect to the considered economies. Allowing for conditional heteroskedasticity within quantiles pays off for the US and EA, while homoskedastic scales are sufficient for the UK. Third, imposing dynamic shrinkage helps predictive accuracy in many cases. The dynamic horseshoe exhibits the most consistent improvements across forecast metrics, horizons and economies. Finally, adjusting for noncrossing quantiles has only minor implications with respect to predictive accuracy.

5.2. Higher-order moments of predictive distributions

To shed light on which features in the resulting predictive distributions yield gains in tail forecast performance, I investigate their shapes in more detail in the following. To recover a smooth estimate of the entire predictive distribution from the individual QRs, I apply kernel smoothing based on a Gaussian kernel on the estimated quantiles (see also [Gaglianone and Lima, 2012](#); [Korobilis, 2017](#)). These distributions are used to generate random samples from the estimated predictive distributions.

In order to compute and investigate higher-order moments, I focus on one-year ahead forecasts for UC-SV, UC-SVM, UCQR-TIS-GPt-dhs and UCQR-TVS-GPt-dhs, and use the following procedure. First, I generate a sample of 3000 random numbers from the respective predictive distribution which I then use to compute the mean, variance, excess kurtosis and skewness of this sample. This step is iterated for 1000 times. Second, to obtain numerical standard errors for the estimates of these moments I compute the 5th and 95th percentile over these 1000 replicates.

The resulting empirical moments are shown in [Fig. 4](#). Starting with the mean of the predictive distribution, several stylized facts across all three economies can be established. While UC-SV and UC-SVM often show similar paths, UC-SVM typically produces more pronounced high-frequency movements in the context of large-variance shocks. This feature is due to feedback effects between inflation and its volatility.

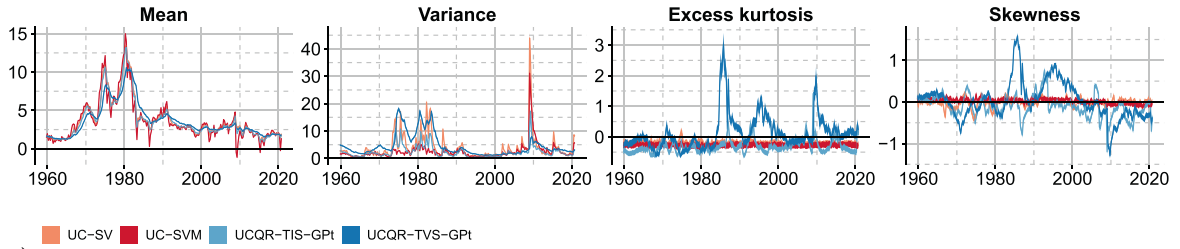
Comparing the two mean-based competing models to the QRs, predictive means from UCQR-TIS often exhibit a similar path, while shifts in the location of the predictive distribution in the case of UCQR-TVS often occur with a delay and are less pronounced. This finding is similar to in-sample evidence, and overall, the predictive mean for the latter model displays a less noisy and smoother evolution. The more aggressive fit of the mean-based models, in light of the forecast results, yields gains in predictive accuracy in the short horizon. However, it appears that a less aggressive fit tends to pay off for higher-order forecasts.

Turning to the variances, several findings are worth noting. In general, time-varying variances for the US and UK are clearly featured for all competing models. This requirement is mostly dictated by the comparatively long sample vis-à-vis the EA spanning different economic phases (see, e.g., [Clark, 2011](#)). By contrast, variances in predictive distributions for the EA are more stable, which corroborates findings by [Jarociński and Lenza \(2018\)](#) and [Huber et al. \(2020\)](#), who detect only limited evidence in favor of stochastic volatility for inflation in the EA. Note that this finding suggests that models for the conditional mean could safely neglect time-varying volatilities, while conditional heteroskedasticity within quantiles via TVS receives empirical support. The UC-SVM model overall yields the narrowest predictive distributions, apart from during the Great Recession in the US. This may be explained by the notion that the volatility process is featured as a predictor, providing a better fit, thereby also reducing the predictive variance. Interestingly, for the US and the EA, predictive variances in normal economic times are typically slightly larger for UCQR-TVS compared to the benchmark models, which can be related to the less aggressive fit with respect to the location of the distribution. During the Great Recession, or the onset of the COVID-19 pandemic for the case of the US, however, the predictive distributions are narrower than those of the mean-based models or UCQR-TIS.

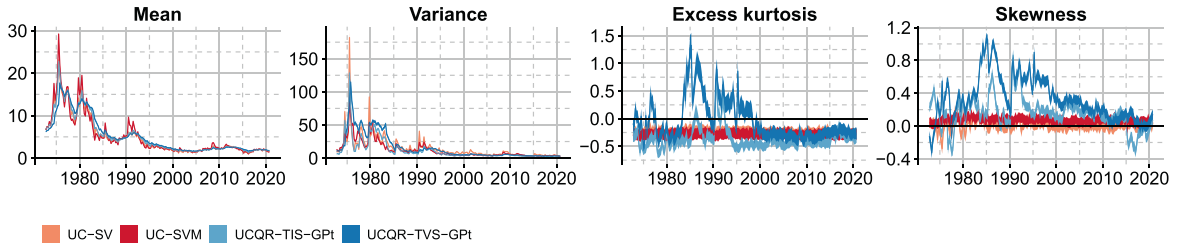
Turning to the higher-order moments – excess kurtosis and skewness – differences between model specifications are more striking. Lower variances during recessionary episodes may in part be explained by noting that particularly UCQR-TVS signals periods of substantial excess kurtosis, pointing to heavier than normal tails of the predictive distribution. Moreover, the resulting distributions are skewed (positively and negatively) during different parts of the samples across economies. Interestingly, UCQR-TIS usually exhibits excess kurtosis values around zero, and less pronouncedly skewed predictive distributions, indicating that this feature arises from allowing for conditional heteroskedasticity within quantiles. Given the comparatively worse performance of UCQR-TVS in the case of the UK, the forecast results suggest that such features are irrelevant to improve density forecast accuracy, while they yield gains for the US and the EA.

It is worth discussing several periods in more detail. Strikingly, excess kurtosis in the predictive distribution for the US occurs after the high-inflation period of the late 1970s. The skewness parameter in conjunction with these heavier than normal tails in the mid 1980s suggests substantial upward inflation risk during the Volcker chairmanship, when inflation expectations were still not properly anchored. Later in the sample, however, during the Great Recession, non-zero excess kurtosis is again measured, but with a negatively skewed distribution, pointing to disinflationary pressures and downward risk. This notion may be linked to the findings of a sign-switch in the relationship between inflation and its volatility identified in [Chan \(2017\)](#). By contrast, the skewness parameter for UCQR-TIS in the UK is positive for most of the sample, pointing mainly towards upward risk early in the sample which has declined since the early 1990s. For the EA, some periods

(a) US



(b) UK



(c) EA

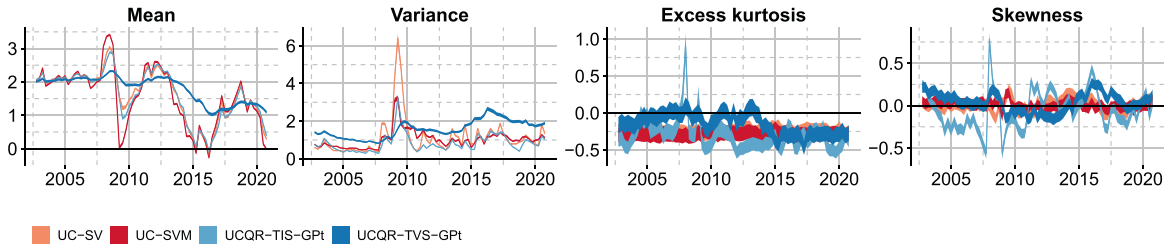


Fig. 4. Higher-order moments of the one-year ahead predictive distributions. *Notes:* Unobserved components model with stochastic volatility (UC-SV), stochastic volatility in mean (UC-SVM) and unobserved components quantile regression (UCQR) with time-invariant (TIS) or time-varying (TVS) scale parameter, adjusted using a Gaussian process (GP) regression with time-varying bandwidth. All models are estimated using the dynamic horseshoe prior. The lines mark numerical 5th and 95th percentiles across 1000 replicates. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

of skewed distributions are featured, particularly during the European debt crisis, but they are not as pronounced as those of the other economies.

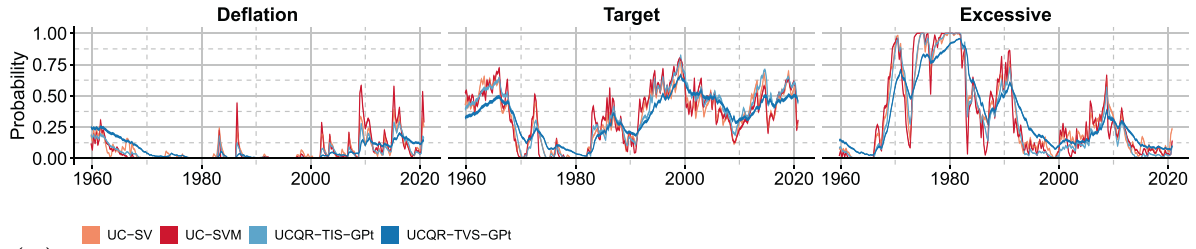
Summarizing, this analysis indicates that forecast gains for the UCQR models can in part be explained by allowing for more flexible predictive distributions featuring both heavier than normal tails and skewness. While such features improve longer-horizon and tail forecasts in the US and the EA, this is not necessarily the case for the UK.

5.3. Tail risks in predictive distributions

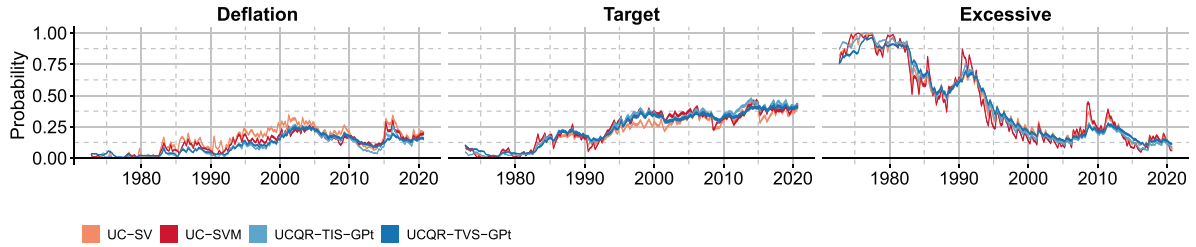
To complement this discussion of tail risks in inflation, and to quantify both upside and downside risk, I compute the probabilities of three different inflation-scenarios: (a) *Deflation* with $\Pr(\pi_{t+h} < 0)$, (b) *Target* with $\Pr(1 \leq \pi_{t+h} \leq 3)$, and (c) *Excessive* with $\Pr(\pi_{t+h} > 4)$. An interesting aspect of this exercise is that while increased variances for UC-SV and UC-SVM usually widen the predictive distribution symmetrically around the mean, this is not necessarily the case for UCQR, which might affect the evolution of scenario probabilities. Moreover, as suggested above, there are several differences between estimates of the predictive mean, variance, excess kurtosis and skewness. All of these may affect the probability of inflation lying within the specified bounds. I again use 1000 replicates of samples from the predictive distributions and compute the numerical 5th and 95th percentiles. The results are shown in Fig. 5.

The previous findings of UCQR-TVS producing smoother moments of predictive distributions carry over to this exercise. Overall, all models tend to agree on the respective probabilities of the considered scenarios, with some differences in the case of the EA. For the US, deflationary risks were virtually non-existent between 1970 and 1985. They appear to be a threat particularly in more recent years, with similarities to the case of the EA. Conversely, probabilities close to one of excessive inflation are visible during the high-inflation periods during the 1980s for the US and the UK, but show overall declines with minor peaks for the rest of the sample, while being only minor in the EA. This is consistent with the overall trend towards

(a) US



(b) UK



(c) EA

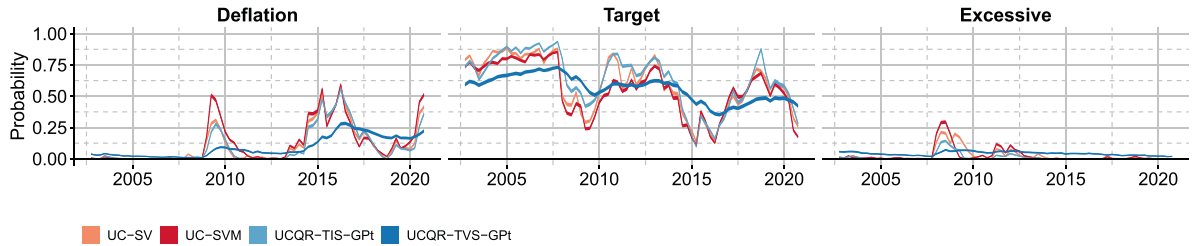


Fig. 5. Predictive one-year ahead scenario probabilities. *Notes:* Unobserved components model with stochastic volatility (UC-SV), stochastic volatility in mean (UC-SVM) and unobserved components quantile regression (UCQR) with time-invariant (TIS) or time-varying (TVS) scale parameter, adjusted using a Gaussian process (GP) regression with time-varying bandwidth. All models are estimated using the dynamic horseshoe prior. Scenarios (a) *Deflation* with $\Pr(\pi_{t+h} < 0)$, (b) *Target* with $\Pr(1 \leq \pi_{t+h} \leq 3)$, and (c) *Excessive* with $\Pr(\pi_{t+h} > 4)$. The lines mark numerical 5th and 95th percentiles across 1000 replicates. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

more stable prices in the last three decades. Notable upward risks occur prior to the early 1990s recession in the US and the ERM crisis in 1992 in the UK; moreover, I detect inflationary risks just before the Great Recession for both economies, and also for the EA.

Turning to the *Target* scenario, which reflects the ability of central banks to keep inflation anchored at the respective target, I find several interesting patterns. Both in the US and the UK, there is a trend towards increasing probabilities of keeping inflation within the interval between one and three percent since the 1980s and the 1990s, respectively. A theoretical argument for this break is provided by noting that the Volcker disinflation in the case of the Federal Reserve, and the adoption of inflation targeting by the Bank of England in 1992, constitutes a policy regime change in the approach to central banking. For a theoretical model discussing breaking the link between inflation and inflation volatility via policy regime changes, and, with it, stabilization of price dynamics, see [Cukierman and Meltzer \(1986\)](#).

Further increasing probabilities of keeping inflation on target in recent years may be linked to more transparent central bank communication and expectation management strategies (see, e.g., [Blinder et al., 2008](#)). Compared to the Federal Reserve and the Bank of England, the European Central Bank exhibits stable and higher probabilities of fulfilling its target measured by UCQR (albeit for a much shorter sampling period). It is worth mentioning that a slight downward trend in target-probabilities is observable for the EA particularly in the aftermath of economic crises, usually coinciding with upward movements in deflationary pressures.

6. Closing remarks

In this paper I discuss methods for estimating Bayesian TVP-QRs featuring conditional heteroskedasticity within quantiles. I rely on data augmentation schemes to render the model conditionally Gaussian and develop an efficient Gibbs sampling algorithm. The high-dimensional parameter space is regularized via dynamic shrinkage priors.

A simple variant of the resulting framework is applied to modeling the distribution of inflation in the US, UK and EA. This model is a quantile-based version of the popular unobserved components model, which has had great success in modeling and forecasting inflation. An out-of-sample forecast horserace indicates that the resulting models are competitive and perform particularly well for higher-order and tail forecasts. The forecast performance indicates that mean-based models are often sufficient for the short-run, but UCQR, particularly equipped with TVS, is capable of improving long-run forecast accuracy considerably. A detailed analysis of the resulting predictive distributions reveals that they are sometimes skewed and feature heavy tails. Investigating implied probabilities of several inflation scenarios reveals that excessive inflation was a threat early in the sample in the US and the UK. Deflationary risks emerged mostly in more recent years.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jedc.2022.104493](https://doi.org/10.1016/j.jedc.2022.104493)

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